Embedded System V.IMP Questions
EduShine Classes – Arman Ali

Unit – 1 Introduction to Embedded System

1. Define an embedded system. What are the key characteristics of embedded systems?(V.V.IMP)
2. Briefly Classify the Embedded System (V.V.IMP)
3. Difference between microcontroller and microprocessor.
4. What are the main hardware components of an embedded system?
5. What are the different types of embedded software? Explain each with examples.
6. What are the steps involved in the development of embedded system software?(V.V.VIMP)
7. Explain Building Process of an Embedded System ?
8. Difference between Embedded System and Real time Embedded System.
9. Explain the major structural units of an embedded processor with their functions.(V.V.IMP)
10. What factors should be considered while selecting a processor for an embedded system?
11. What are the different types of memory devices used in embedded systems? Explain with examples.(V.VIMP)
12. What is Direct Memory Access (DMA) and how does it it Works in Embedded System?(V.VIMP)
13. Explain memory management methods in embedded systems and why are they important?(V.VIMP)
14. Explain the Following :

    i. Timer and Counter
    ii. Watchdog Timer
    iii. Real Time Clock(RTC)
    iv. ICE & Target hardware Debugging

Unit – 2 Embedded Networking

1. Define Embedded Networking. What types of Networks Used in Embedded System?(IMP)
2. Discuss the types of Networking Protocol used in Embedded System.(V.IMP)
3. What are I/O device ports in embedded systems? Explain their types and importance.
4. What is Data Communication in Embedded System. Why it is Needed ?
5. Explain Serial Bus Communication and its transmission mode (Simplex,half duplex, Full Duplex).(V.VIMP)
6. Discuss Serial bus Communication Protocol : (V.VIMP)

   i. RS232
   ii. RS422
   iii. RS485

7. Differentiate between synchronous and asynchronous data transmission with examples.
8. What is the CAN Bus protocol? Explain its working and applications in embedded systems.
9. What is SPI communication protocol? How does it work in embedded systems?(V.IMP)
10. What is I2C communication? How is it different from SPI?(V.IMP)
11. Why are device drivers important in embedded systems? Explain with examples.
12. Discuss about the application of Embedded System & its design issue ?
13. Define Sensor and Actuators. Explain Briefly device driver.
14. What is Task Synchronization in Embedded Systems. Explain its Techniques.
15. Write Short Note on :

   i. USB ii. Firewire iii. Bluetooth iv. ZigBee

Unit – 3 Embedded Firmware Development Environment

1. What is EDLC in embedded systems? What are its main objectives?.(V.VIMP)
2. What are the different phases of the Embedded Product Development Life Cycle (EDLC)? Explain each phase briefly.(V.VIMP)
3. What is the Waterfall Model in Embedded System Development Life Cycle (EDLC)? Explain its stages and advantages.(V.IMP)
4. What is the Iterative Model in EDLC? How does it differ from the Waterfall Model?(V.VIMP)
5. What is the Prototype Model in EDLC? Explain its key features and when it is used in embedded systems development.(V.IMP)
6. What is the Spiral Model in EDLC? How does it combine aspects of both design and prototyping?
7. What is Hardware Software Co-Design. Discuss 4 Common Issues in Hardware-Software Co-Design.(IMP)
8. What is a data flow graph model? How is it used in embedded system design?
9. What is a state machine model? Explain its role in embedded system behavior representation.(V.IMP)
10. What is a concurrent process model? How does it help in multitasking in embedded systems?(V.VIMP)
11. What is a sequential program model in embedded systems? Where is it commonly used?
12. What is an object-oriented model in embedded systems? Explain its benefits in system design.

Unit – 4 RTOS Based Embedded System Design

1. Explain Real Time Operating System & its types. Explain also its Key features.(V.VIMP)
2. Difference between Hard RTOS and Soft RTOS.(IMP)
3. Discuss Characteristics of Real Time Operating System(RTOS).(V.IMP)
4. Difference between Process and Thread.
5. Explain Life Cycle of Process/Task.(V.IMP)
6. What is ISR(Interrupt Service Routine). How Interrupt is Handled in RTOS?(V.IMP)
7. Difference between Multitasking and Multiprocessing.
8. What is context switching in embedded systems or operating systems? Explain with an example.
9. What is the difference between preemptive and non-preemptive scheduling in embedded systems?
10. What is the role of synchronization in inter-process communication? Discuss Task Synchronization Techniques.(V.IMP)
11. What is priority inversion, and why is it a problem in real-time systems?
12. How does priority inheritance help to resolve priority inversion issues?
13. Compare the ease of use and system overhead between VxWorks, µC/OS-II, and RT Linux.(V.VIMP)

Unit – 5 Embedded System Application Development

1. Discuss Design Issues and Techniques in Embedded System Application Development.(V.VIMP)
2. Discuss the Case study of Washing Machine.(V.V.VIMP)
3. Discuss the Case Study of Automotive Applications in Embedded Systems.
4. Discuss Case Study: Smart Card System Application.(V.V.VIMP)